

Karthik Valmeekam

✉ kvalmeek@asu.edu
📁 [karthikv792.github.io](https://github.com/karthikv792)
🌐 [karthikv792](https://www.linkedin.com/in/karthikv792)
🎓 Scholar profile

Research Interests

My research sits at the intersection of Large Language Models (LLMs) and reasoning, with a focus on planning, along three threads:

- exposing the limits of LLMs in autonomous planning, showing their apparent abilities stem from pattern matching rather than robust reasoning;
- designing architectures that reposition LLMs as idea generators within verified pipelines with correctness guarantees; and
- interrogating the intermediate “reasoning” of Large Reasoning Models, showing that models trained on semantically meaningless traces can match or surpass those trained on correct ones, challenging the anthropomorphization of these tokens as “thoughts.”

This work has been recognized with the **IBM PhD Fellowship** (2024), the **ICAPS Distinguished Dissertation Award** (2026), and ASU's **Dean's Dissertation Award** (2026).

Experience

- June 2025 – Present **amazon AGI**, *Applied Scientist II*, Sunnyvale, CA, United States.
May 2024 – August 2024 **amazon AGI**, *Applied Scientist Intern*, Palo Alto, CA, United States.

Education

- Fall 2021 – Spring 2026 **Arizona State University**, Tempe, AZ, United States
Ph.D. in Computer Science, Advisor: *Prof. Subbarao Kambhampati*
Dissertation: On the Role Of Large Language Models in Planning
ICAPS Distinguished Dissertation Award; Dean's Dissertation Award
GPA: 4.0/4.0
- Fall 2019 – Spring 2021 **Arizona State University**, Tempe, AZ, United States
M.S. in Computer Science, Advisor: *Prof. Subbarao Kambhampati*
Thesis: A Study of Explainable Decision Support for Longitudinal Sequential Decision Making
GPA: 4.0/4.0
- Fall 2015 – Spring 2019 **Vellore Institute of Technology**, Vellore, TN, India
B.Tech. in Computer Science and Engineering
GPA: 9.19/10.0

Publications

As of Mar 19th 2026, per google scholar, my papers have received a total of 3595 citations; I have an h-index of 18 and an i10-index of 20.

- TMLR 2026 **Karthik Valmeekam***, Vardhan Palod*, Kaya Stechly* (equal contribution), Atharva Gundawar, Subbarao Kambhampati. Beyond Semantics: The Unreasonable Effectiveness of Reasonless Intermediate Tokens. In NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning
- ICML 2026 Subbarao Kambhampati, **Karthik Valmeekam**, Siddhant Bhambri, Vardhan Palod, Lucas Saldyt, Kaya Stechly, Soumya Rani Samineni, Durgesh Kalwar, Upasana Biswas. Position: Stop Anthropomorphizing Intermediate Tokens as Reasoning/Thinking Traces!. Annals of the New York Academy of Sciences, 2025.
- NYAS 2025 Subbarao Kambhampati, Kaya Stechly, **Karthik Valmeekam**. (How) Do reasoning models reason?. Annals of the New York Academy of Sciences, 2025.
- TMLR 2025 **Karthik Valmeekam***, Kaya Stechly* (equal contribution), Atharva Gundawar, Subbarao Kambhampati. A Systematic Evaluation of the Planning and Scheduling Abilities of the Reasoning Model o1. Transactions on Machine Learning Research, 2025.
- ICLR 2025 Kaya Stechly*, **Karthik Valmeekam*** (equal contribution), Subbarao Kambhampati. On the self-verification limitations of large language models on reasoning and planning tasks. arXiv preprint arXiv:2402.08115, 2024.
- NeurIPS 2024 Kaya Stechly*, **Karthik Valmeekam*** (equal contribution), Subbarao Kambhampati. Chain of thoughtlessness: An analysis of cot in planning. arXiv preprint arXiv:2405.04776, 2024.
- ICML 2024 Subbarao Kambhampati, **Karthik Valmeekam**, Lin Guan, Kaya Stechly, (Spotlight) Mudit Verma, Siddhant Bhambri, Lucas Saldyt, Anil Murthy. LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks. In Forty-first International Conference on Machine Learning, 2024.
- AAAI 2024 Subbarao Kambhampati, **Karthik Valmeekam**, Lin Guan. On the role of large language models in planning. Tutorial at the AAAI Conference on Artificial Intelligence, 2024.
- NeurIPS 2023 **Karthik Valmeekam**, Matthew Marquez, Sarath Sreedharan, Subbarao Kambhampati. On the Planning Abilities of Large Language Models—A Critical Investigation. In Thirty-seventh Conference on Neural Information Processing Systems, 2023.
Spotlight paper (Top 3%).
- NeurIPS 2023 Lin Guan*, **Karthik Valmeekam*** (equal contribution), Sarath Sreedharan, Subbarao Kambhampati. Leveraging Pre-trained Large Language Models to Construct and Utilize World Models for Model-based Task Planning. In Thirty-seventh Conference on Neural Information Processing Systems, 2023.

- NeurIPS 2023 **Karthik Valmeekam**, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, Subbarao Kambhampati. PlanBench: An Extensible Benchmark for Evaluating Large Language Models on Planning and Reasoning about Change. In Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2023.
- FMDM @ NeurIPS 2023 **Karthik Valmeekam***, Matthew Marquez* (equal contribution), Subbarao Kambhampati. Can Large Language Models Really Improve by Self-critiquing Their Own Plans?. In NeurIPS 2023 Foundation Models for Decision Making Workshop, 2023.
- ICAPS 2023 Subbarao Kambhampati, **Karthik Valmeekam**, Matthew Marquez, Lin Guan. On the role of large language models in planning. Tutorial at the International Conference on Automated Planning and Scheduling (ICAPS), 2023.
- ICLR 2023 Lin Guan, **Karthik Valmeekam**, Subbarao Kambhampati. Relative Behavioral Attributes: Filling the Gap between Symbolic Goal Specification and Reward Learning from Human Preferences. In The Eleventh International Conference on Learning Representations, 2023.
- ICAPS 2022 **Karthik Valmeekam**, Sarath Sreedharan, Sailik Sengupta, Subbarao Kambhampati. RADAR-X: An Interactive Mixed Initiative Planning Interface Pairing Contrastive Explanations and Revised Plan Suggestions. In Proceedings of the International Conference on Automated Planning and Scheduling (Vol. 32, pp. 508-517), 2022.

Teaching & Service

- Teaching Assistant 2022: **CSE574-Planning and Learning** by Prof. Subbarao Kambhampati.
2021: **CSE471-Intro to Artificial Intelligence** by Prof. Subbarao Kambhampati.
- Reviewer/PCM 2026: **ICAPS**
2025: **JAIR, ICLR, ACM-TIST, CACM, AAAI-Student Program**.
2024: **ICLR, NeurIPS, AAAI-Student Program, ICRA**.
2023: **ICLR, NeurIPS, ICAPS, HAXP@ICAPS, GenPlan@NeurIPS**.
2022: **XAIP@ICAPS**.
- Student Volunteer 2021: **AAAI, ICAPS**.

Skills

- Programming Python, C++, HTML/CSS, JavaScript, Bash, SQL, PDDL
- Frameworks DeepSpeed ZeRO training, verl (GRPO and FSDP SFT), Pytorch, Transformers (HuggingFace), Scikit-Learn, Numpy, Pandas, Flask, JQuery, Bootstrap
- Misc Adobe Premiere Pro, Lightroom, Photoshop, Git, Latex

Projects

- 2021 **Reinforcement Learning for Imperfect Information Games**
On a two-player imperfect information game (Sequence), an agent was trained offline by *approximation in value and policy space* using neural networks. In the test phase, the agent used one-step look-ahead using the trained network to make a move. This improved win percentage (upto 60%) with more training against a random agent.
- 2020 **High-Speed Autonomous Drifting using Deep Reinforcement Learning**
An end-to-end drift controller which utilizes a state-of-the-art model-free reinforcement learning algorithm is modelled using *Pytorch*. This RL problem is devised as a trajectory following problem. The resulting agent was able to generate the required behavior.

Honors

- 2024-2025 IBM PhD Fellowship Award
- 2025 SCAI Doctoral Fellowship (Merit-Based), Arizona State University.
- 2024 SCAI Doctoral Fellowship (Merit-Based), Arizona State University.
- 2021 CIDSE Doctoral Fellowship (Merit-Based), Arizona State University.
- 2018 Vice Chair Management of the student chapter CODECHEF-VIT, Vellore Institute of Technology